

Topology PhD Exam, September 1989

1. State the exact homology sequence of a pair.
2. Describe all coverings of the torus $T^2 = S^1 \times S^1$.
3. State the Lefschetz fixed point theorem for compact simplicial complexes.
4. State Tietze's extension theorem for normal spaces.
5. Let $K = K_1 \cup K_2$ be a simplicial complex with K_1 and K_2 subcomplexes, and let $A = K_1 \cap K_2$. State the Mayer–Vietoris sequence for this.
6. Let X be the set of points in the plane which is the closure of the set $S = \{(x, \sin(1/x)) \mid 0 < x \leq 1\}$. Show that there is no continuous mapping $f : [0, 1] \rightarrow X$ which is onto.
7. Prove that a compact Hausdorff space is normal.
8. Let $f, g : X \rightarrow S^n$ be any continuous mappings. Suppose that for all $x \in X$, $f(x)$ and $g(x)$ are not antipodal. Show that f and g are homotopic.
9. Use the van Kampen theorem to compute the fundamental group of the surface of genus 2.
10. Give an example of a space X for which $\pi_1(X, x_0)$ is not finitely generated.